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Iwamoto et al.

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(54) **MICROSCOPE UNIT**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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5,325,231 A * 6/1994 Tamura G02B 21/088
359/387
5,619,031 A * 4/1997 Choate G02B 27/34
348/136

(Continued)

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FOREIGN PATENT DOCUMENTS

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EP 3 367 152 A1 8/2018
JP 50-124980 U 10/1975
(Continued)

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OTHER PUBLICATIONS

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Primary Examiner — Jonathan Y Jung

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G02B 21/08 (2006.01)
G02B 21/36 (2006.01)
G02B 27/10 (2006.01)

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21/361 (2013.01); **G02B 21/362** (2013.01);
G02B 27/1013 (2013.01)

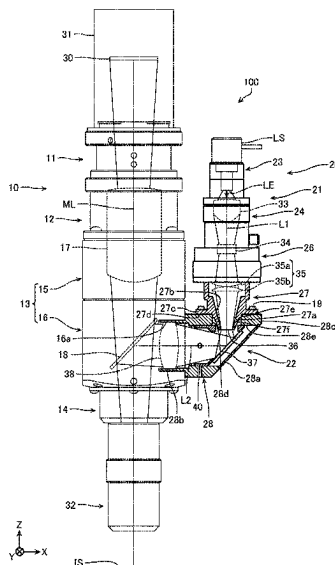
(58) **Field of Classification Search**

CPC G02B 7/002; G02B 7/02–16; G02B
21/00–368; G02B 25/00; G02B 25/001
See application file for complete search history.

(57) **ABSTRACT**

A microscope unit comprises: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source, the illumination lens barrel having: a first lens barrel configured capable of being fitted with the light source; and an intermediate lens barrel connecting the main lens barrel and the first lens barrel, and a field stop of light irradiated from the light source being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel is.

12 Claims, 8 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

6,507,434	B2	1/2003	Miyashita	
7,576,928	B2 *	8/2009	Tanabe	G02B 21/33 359/813
2010/0141939	A1 *	6/2010	Zhan	G01Q 30/02 850/52
2018/0231755	A1 *	8/2018	Miyake	G02B 27/0961
2020/0126752	A1 *	4/2020	Brodie	H01J 37/12
2022/0382033	A1	12/2022	Honma et al.	

FOREIGN PATENT DOCUMENTS

JP	2002-6225	1/2002
JP	1708088 S	2/2022
JP	2022-181144 A	12/2022
WO	2017/068627 A1	4/2017

* cited by examiner

FIG. 1

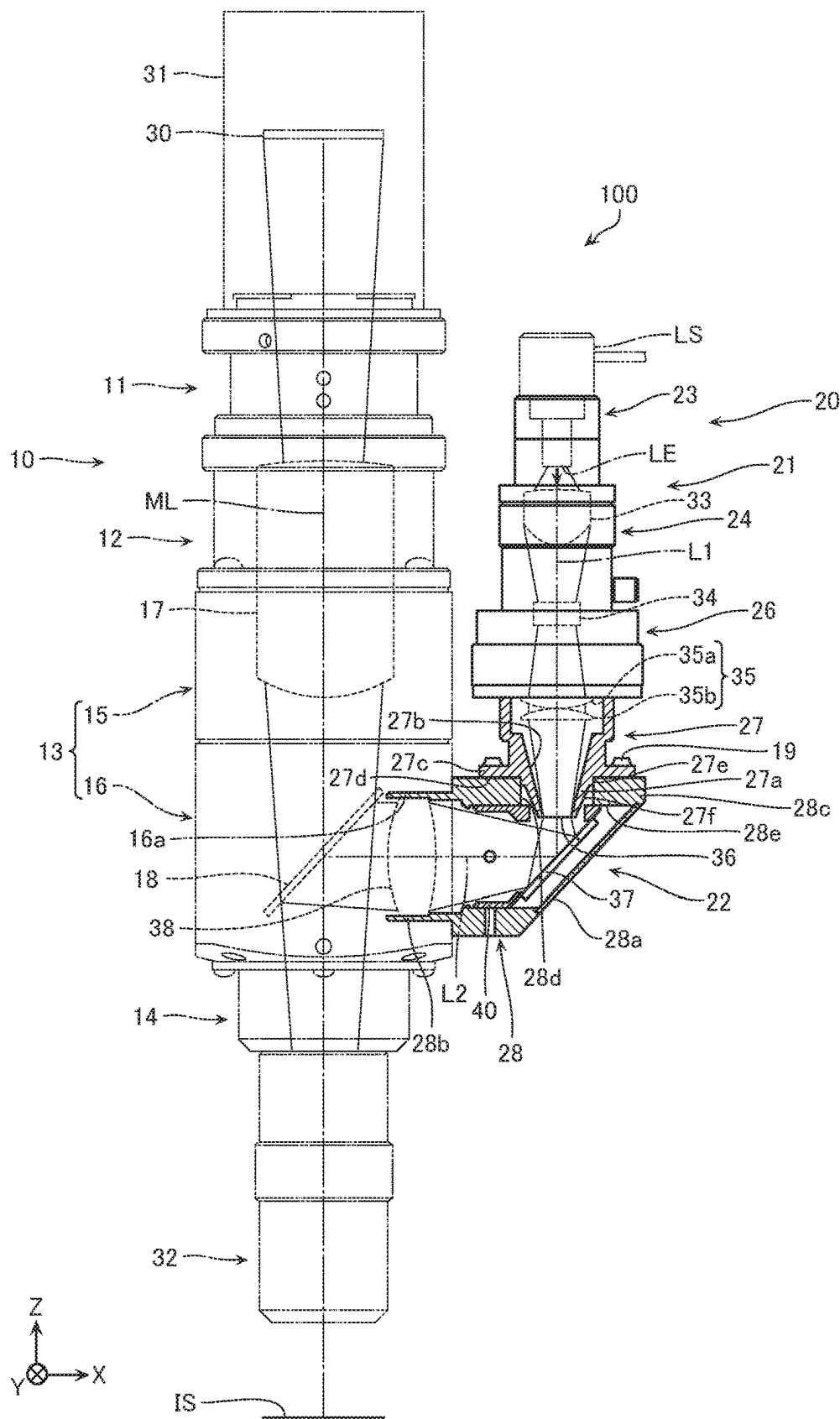


FIG. 2

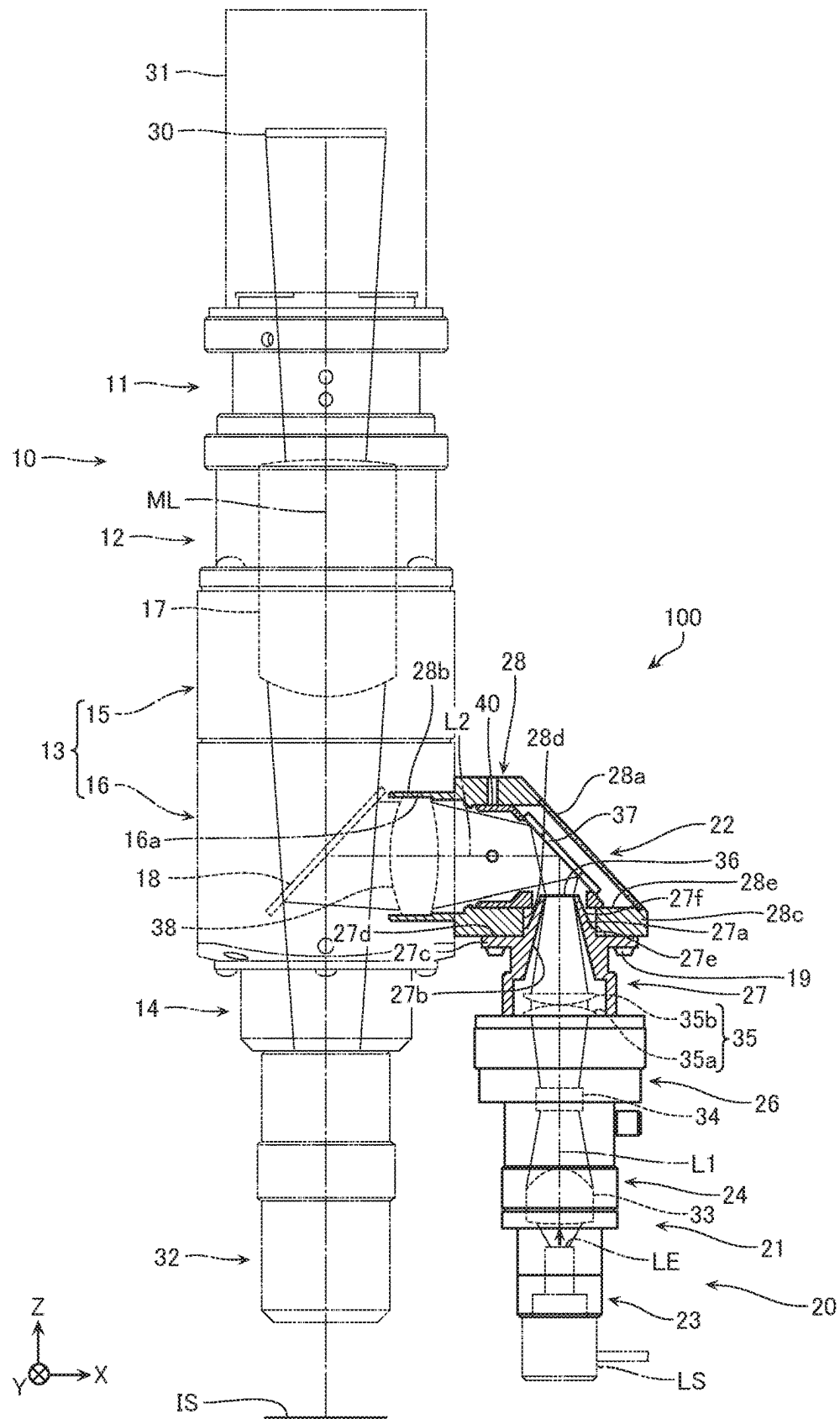


FIG. 3

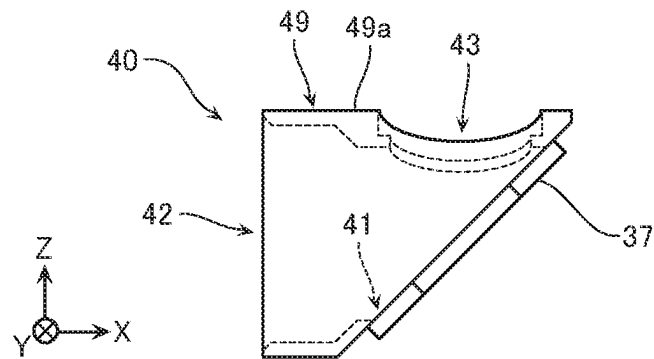


FIG. 4

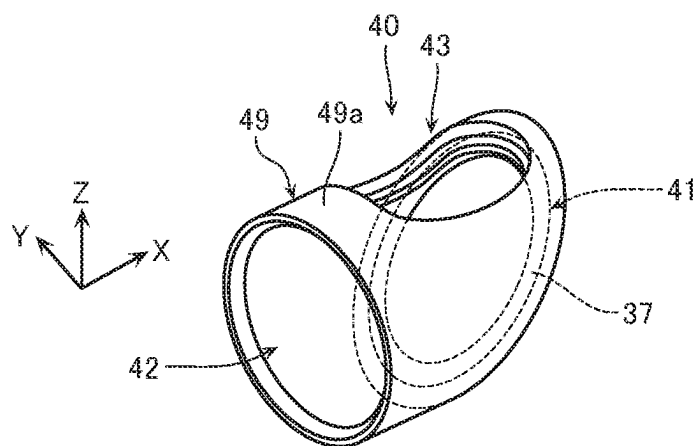


FIG. 5a

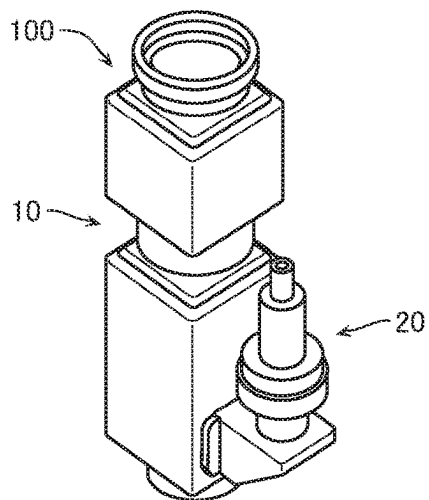


FIG. 5b

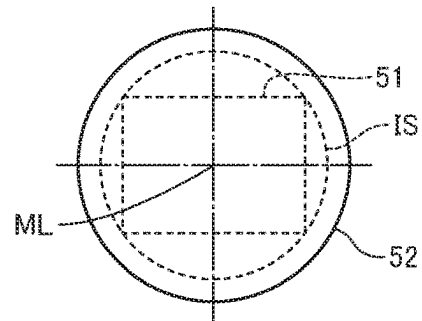


FIG. 6a

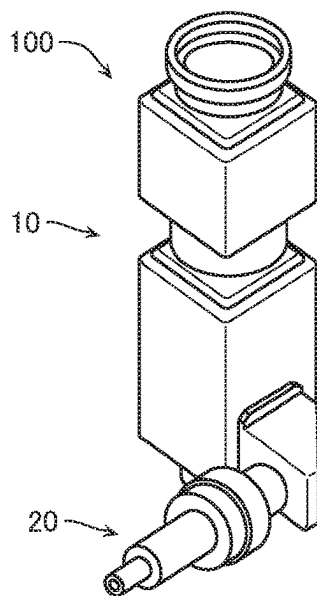


FIG. 6b

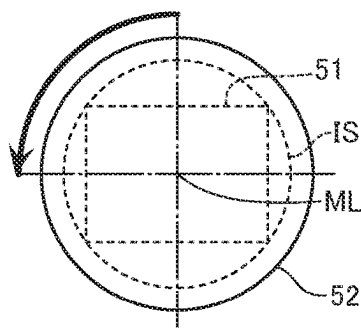


FIG. 6c

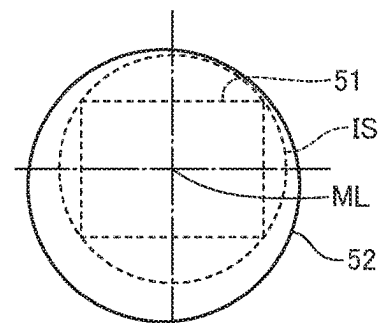


FIG. 7a

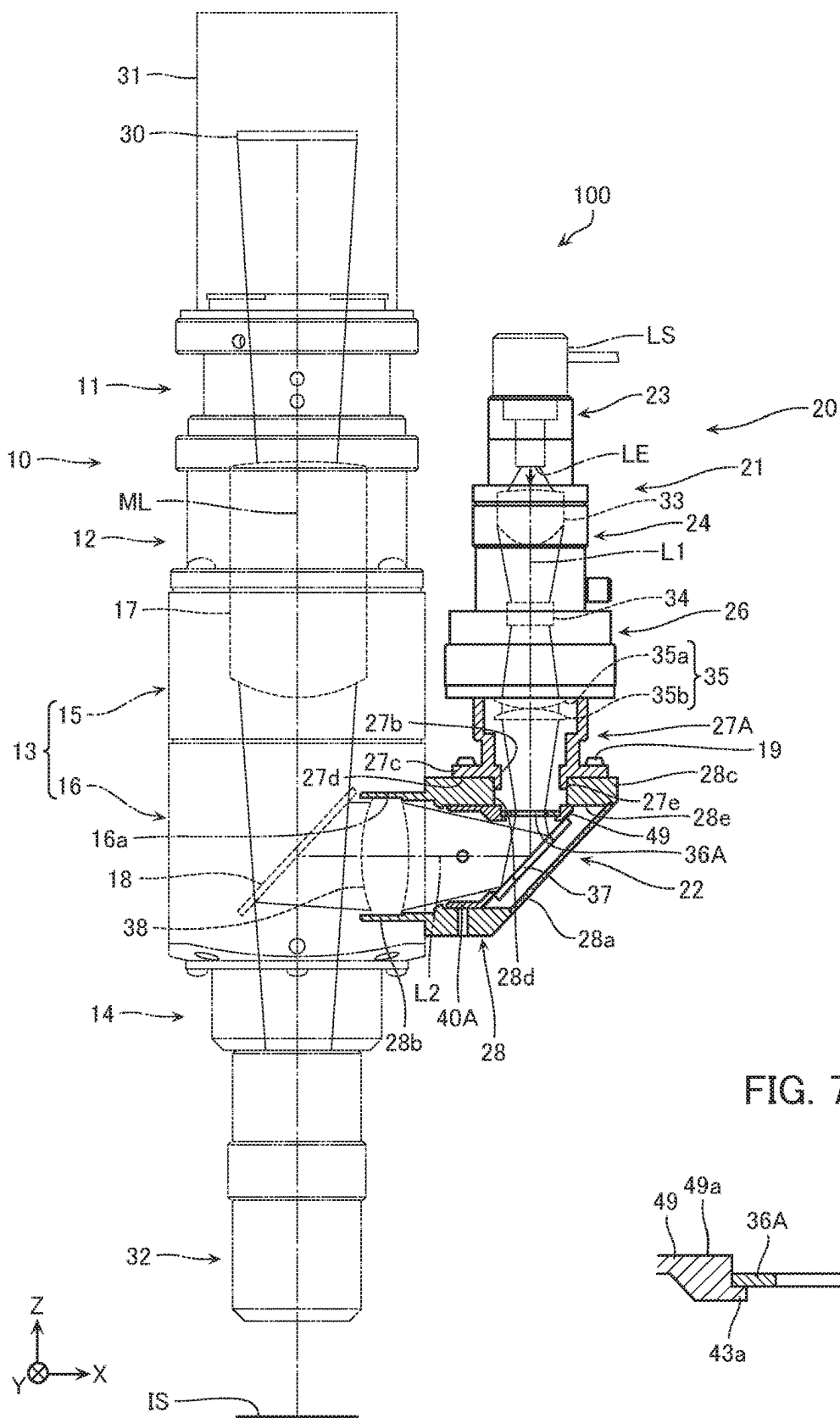


FIG. 7b

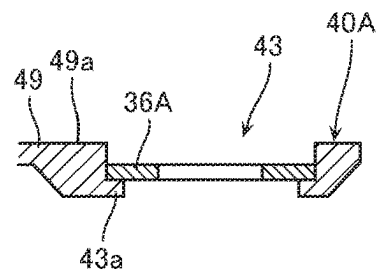


FIG. 8a

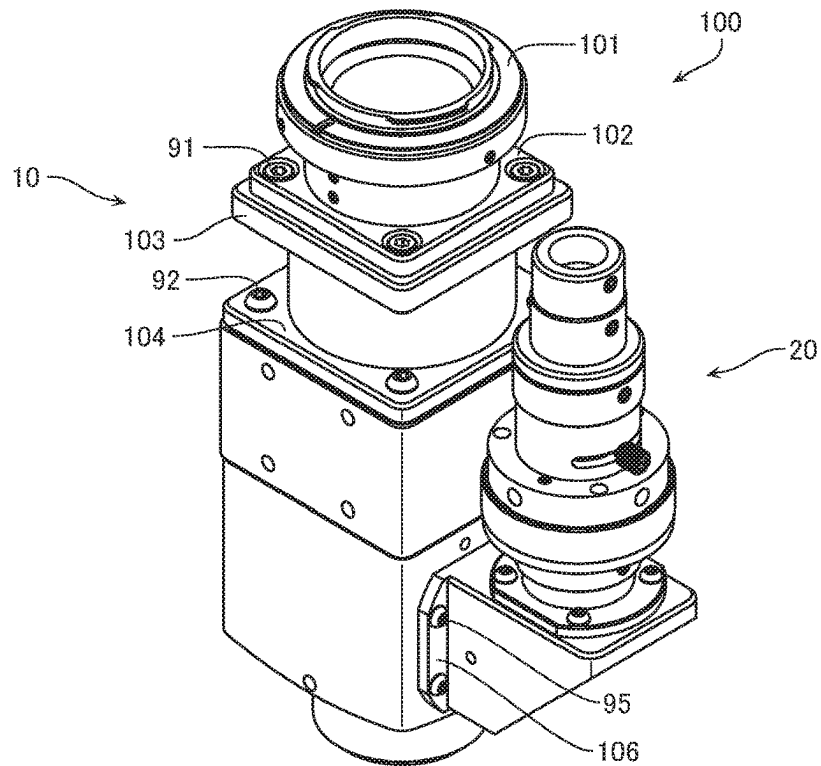
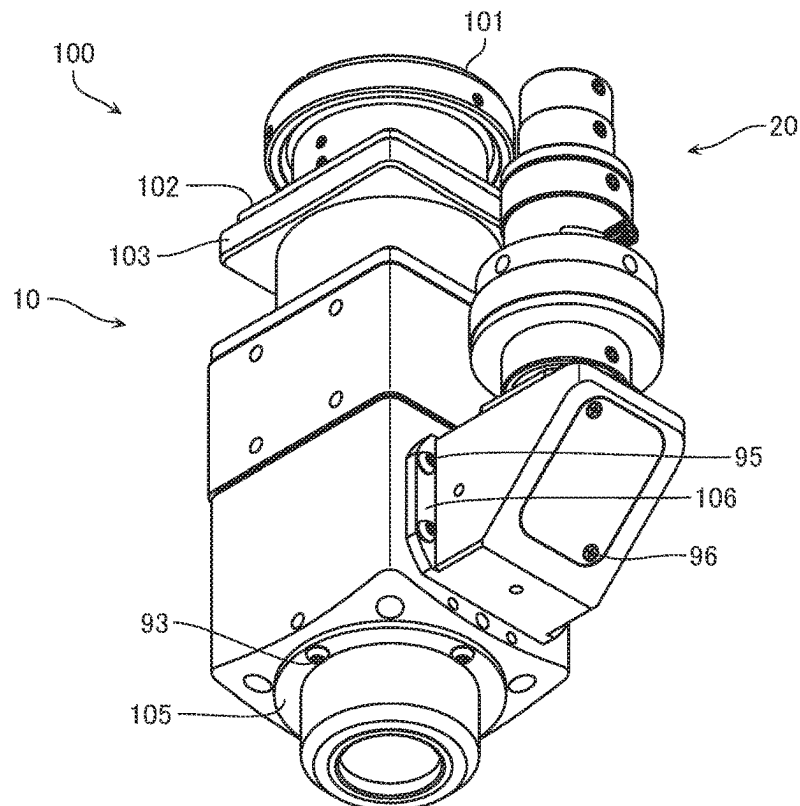


FIG. 8b



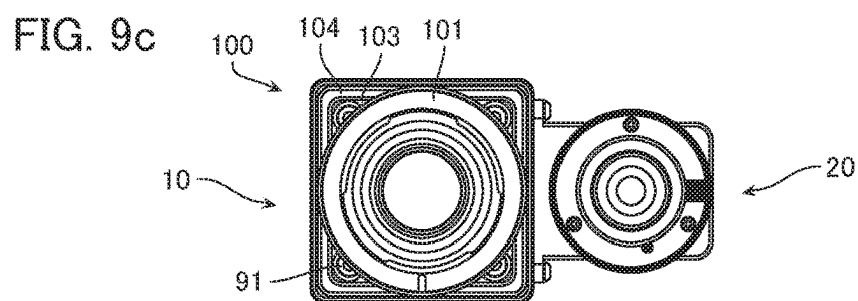
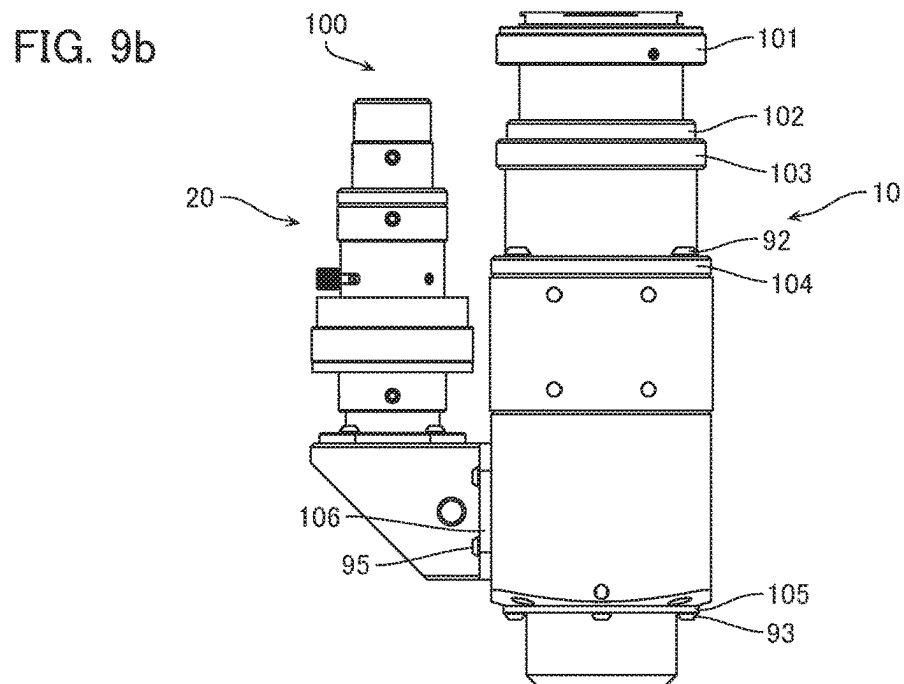
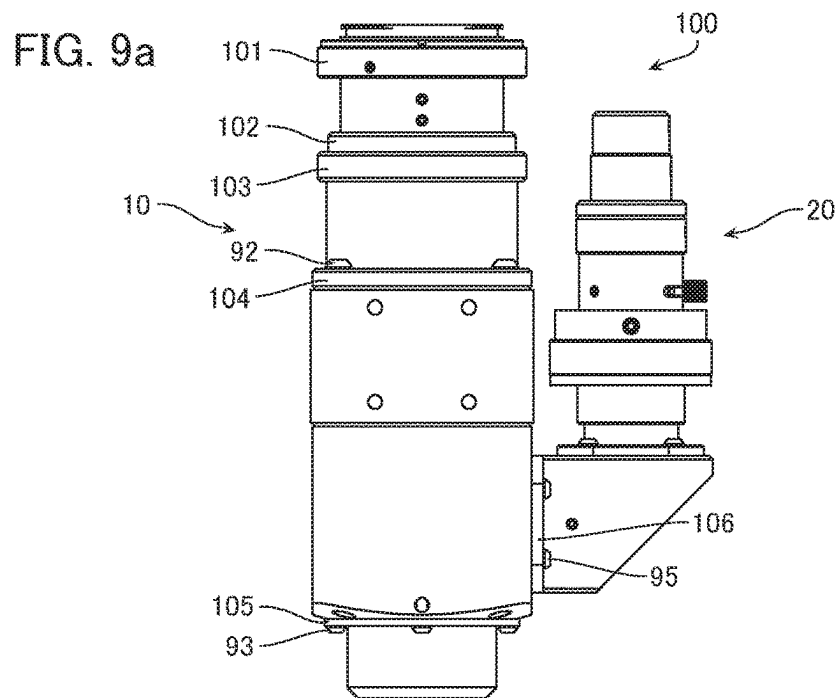


FIG. 10a

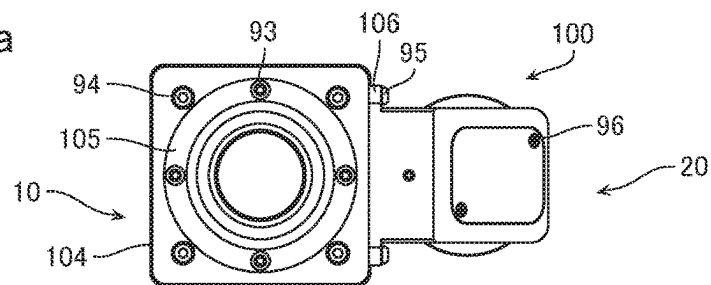


FIG. 10b

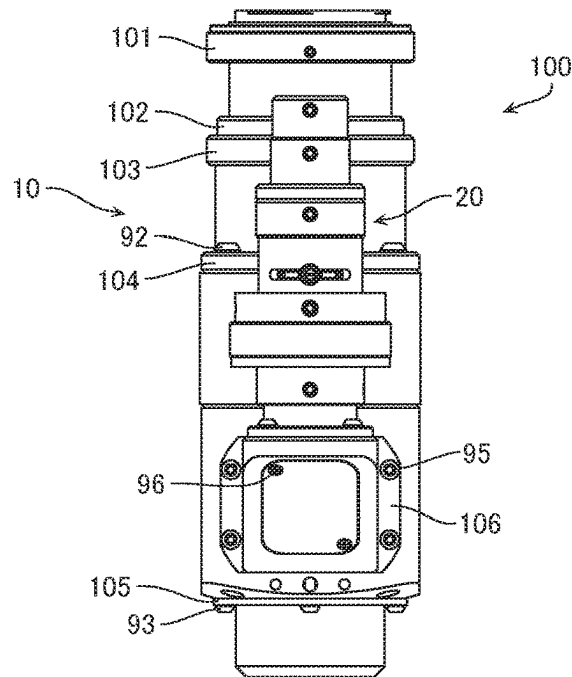
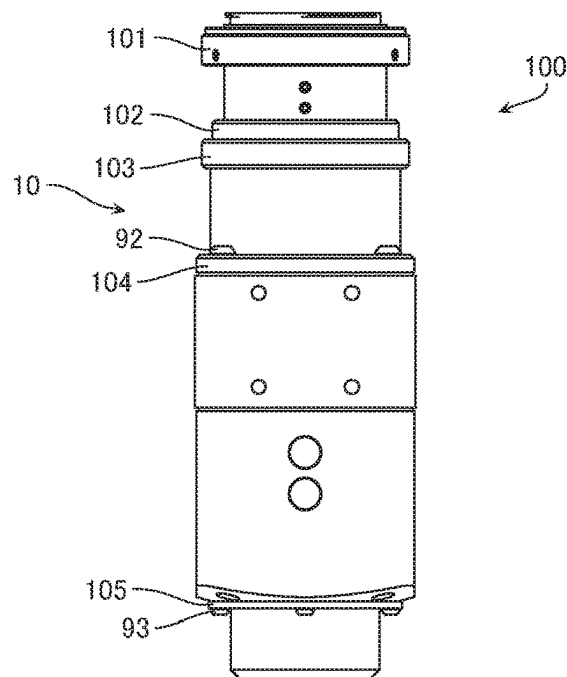


FIG. 10c



1 MICROSCOPE UNIT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is based upon and claims the benefit of Japanese Patent Application No. 2021-088019, filed on May 25, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

Field

The present invention relates to a microscope unit capable of being fitted to a target device.

Description of the Related Art

Up to now, there has been known a microscope unit comprising: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source. Such a microscope unit is configured so that, for example, a light flux (a flux of diverging light rays) that has been emitted from the light source is collected by a collector lens, passes through a relay lens system and field stop to be reflected by a mirror, and passes through a relay lens system, aperture stop, and condenser lens to illuminate an observation surface.

Such a microscope unit is sometimes capable of being fitted with a variety of light sources. However, it has sometimes been the case that, depending on a kind of light source fitted to the microscope unit, illumination at the observation surface ends up getting ununiform. Accordingly, for example, in a microscope illumination device described in Japanese Patent Application Publication No. 2002-6225, a fly-eye lens is disposed in the illumination optical system, whereby uniformity of illumination at the observation surface is achieved.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic explanatory diagram showing overall configuration of a microscope unit according to a first embodiment.

FIG. 2 is a schematic explanatory diagram showing overall configuration of the microscope unit.

FIG. 3 is a schematic side view showing configuration of a mirror frame member of the microscope unit.

FIG. 4 is a schematic perspective view showing configuration of the mirror frame member of the microscope unit.

FIGS. 5a and 5b are diagrams for explaining a relationship between attitude of the illumination lens barrel with respect to the main lens barrel of the microscope unit and illumination range.

FIGS. 6a, 6b and 6c are diagrams for explaining illumination range in the case of fitting position adjustment having been completed and the case of fitting position adjustment not yet having been performed.

FIGS. 7a and 7b are schematic explanatory diagrams showing configuration of a microscope unit according to a second embodiment.

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FIGS. 8a and 8b are explanatory diagrams showing outward appearance of the microscope unit.

FIGS. 9a, 9b and 9c are explanatory diagrams showing outward appearance of the microscope unit.

FIGS. 10a, 10b and 10c are explanatory diagrams showing outward appearance of the microscope unit.

DETAILED DESCRIPTION

In the above-described microscope illumination device, the fly-eye lens needs to be provided in the illumination optical system. Moreover, a relay lens needs to be provided between the fly-eye lens and the field stop in order to relay light from the fly-eye lens. There is a risk that this will end up leading to an increase in size of the illumination lens barrel and, consequently, end up leading to an increase in size of the microscope unit.

The present invention, which was made in view of such a problem, has an object of providing a microscope unit that enables downsizing of an illumination lens barrel overall to be achieved, and, consequently, enables downsizing of the microscope unit to be achieved.

A microscope unit according to one embodiment of the present invention comprises: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source, the illumination lens barrel having: a first lens barrel configured capable of being fitted with the light source; and an intermediate lens barrel connecting the main lens barrel and the first lens barrel, and a field stop of light irradiated from the light source being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel is.

In this kind of microscope unit, the field stop of the illumination lens barrel is disposed more inwardly than the outer peripheral surface of the intermediate lens barrel is. Hence, it is possible for space inside the first lens barrel to be utilized more effectively compared to in such a case as when, for example, the field stop is disposed more outwardly than the outer peripheral surface of the intermediate lens barrel is. This makes it possible for total length of the illumination lens barrel to be configured short, whereby downsizing of the illumination lens barrel overall is achieved, and, consequently, downsizing of the microscope unit is achieved.

The above-described field stop may be disposed more inwardly than an inner peripheral surface of the intermediate lens barrel is.

The above-described intermediate lens barrel may be connected to the main lens barrel so as to freely revolve around its own optical axis with respect to the main lens barrel.

The above-described microscope unit may have: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel; a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member. Moreover, in such a case, the field stop may be disposed more inwardly than an outer peripheral surface of the mirror frame member is.

Moreover, the illumination optical system may have: a collector lens that collects light that has been irradiated from the light source; a fly-eye lens allowing to be transmitted

therethrough light from the collector lens; a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens; the reflecting mirror that reflects light from the first relay lens toward the main lens barrel; a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and a second relay lens that relays to the beam splitter light that has been reflected by the reflecting mirror. In such a case, there may be adopted a configuration of a kind that the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens, the intermediate lens barrel houses the second relay lens, and the main lens barrel houses the beam splitter.

Moreover, the field stop may be part of the first lens barrel, and may be housed in the intermediate lens barrel.

Such a configuration makes it possible to provide a microscope unit that enables downsizing of an illumination lens barrel overall to be achieved, and, consequently, enables downsizing of the microscope unit to be achieved.

Microscope units according to several embodiments of the present invention will be described in detail below with reference to the accompanying drawings. However, the embodiments below are not to limit the inventions according to each of the claims, moreover, not all of the combinations of features described in the embodiments are necessarily essential to the means for solving the problem of the invention. Note that in the embodiments below, scales, dimensions, and so on of each of configuring elements are sometimes shown expanded, and some of the configuring elements are sometimes omitted.

Moreover, in the description below, an "X axis direction" means a left-right direction in the case of having correctly faced the paper surface (front surface) of a microscope unit shown in FIG. 1, a "Y axis direction" means a depth direction in same case, and a "Z axis direction" means an up-down direction, that is, an optical axis direction of a main lens barrel in same case. Note that basic structure of a microscope unit is already known, so, apart from where necessary, overall configuration will be described here in outline only.

First Embodiment

[Configuration]

FIGS. 1 and 2 are schematic explanatory diagrams showing overall configuration of a microscope unit 100 according to a first embodiment. In FIGS. 1 and 2, part of the microscope unit 100 is shown as a cross-sectional view, and part of the microscope unit 100 is shown as a transparency view. FIG. 2 shows a situation where attitude with respect to a main lens barrel 10 of an illumination lens barrel 20 differs by 180° compared to in FIG. 1. Hereafter, description will be made assuming a state shown in FIG. 1 to be a basic attitude of the microscope unit 100.

As shown in FIGS. 1 and 2, the microscope unit 100, which configures part of an unillustrated microscope, comprises: the main lens barrel 10; and the illumination lens barrel 20 which is connected to the main lens barrel 10.

The main lens barrel 10, which configures a so-called imaging optical system of the microscope, is for example configured so that its end portion on an upper side in the Z axis direction exemplified in FIG. 1 (hereafter, simply called "upper side") is capable of being fitted with a camera 31 incorporating an imaging sensor 30, and its end portion on

a lower side in the Z axis direction exemplified in FIG. 1 (hereafter, simply called "lower side") is capable of being fitted with an objective lens 32. Note that the imaging sensor 30 includes various kinds of image sensors (CMOS, CCD), and so on.

The main lens barrel 10 has: a first sleeve portion 11 of cylindrical shape disposed on the upper side (a camera 31 side); and a second sleeve portion 12 of cylindrical shape disposed on a lower side of this first sleeve portion 11. The first sleeve portion 11 is formed so that its minimum outer diameter will be slightly smaller than a minimum outer diameter of the second sleeve portion 12.

Moreover, the main lens barrel 10 has: a case of main body portion 13 of rectangular-shaped outward appearance disposed on a lower side of the second sleeve portion 12; and a third sleeve portion 14 of cylindrical shape disposed on a lower side of this case of main body portion 13 and being connected with the objective lens 32. Note that the case of main body portion 13 is configured by: a first casing 15; and a second casing 16 which is disposed on a lower side of this first casing 15.

On insides of the second sleeve portion 12 and the first casing 15 of the case of main body portion 13 in the main lens barrel 10, there is housed an imaging lens (a tube lens) 17. Moreover, on an inside of the second casing 16 of the case of main body portion 13 in the main lens barrel 10, a beam splitter (hereafter, written as "B/S") 18 that will be mentioned later is housed so as to incline at a certain angle to an optical axis ML of the main lens barrel 10. These first sleeve portion 11, second sleeve portion 12, case of main body portion 13, and third sleeve portion 14 configuring the main lens barrel 10 are coaxially disposed linearly in the Z axis direction, along the optical axis ML of the main lens barrel 10.

The illumination lens barrel 20, which configures a so-called illumination optical system of the microscope, has: a first lens barrel 21 capable of having a variety of light sources LS fitted in a freely attachable/detachable manner to its end portion on the upper side in the Z axis direction in the basic attitude; and an intermediate lens barrel 22 that connects this first lens barrel 21 to the main lens barrel 10. Note that the first lens barrel 21 has its optical axis L1 formed in the Z axis direction, and the intermediate lens barrel 22 has its optical axis L2 formed in the X axis direction.

The first lens barrel 21 has: a fourth sleeve portion 23 of cylindrical shape disposed on the upper side in the basic attitude; and a fifth sleeve portion 24 of cylindrical shape disposed on a lower side of this fourth sleeve portion 23. Note that the fourth sleeve portion 23 is formed so that its minimum outer diameter will be smaller than a minimum outer diameter of the fifth sleeve portion 24.

Moreover, the first lens barrel 21 has: a coupler case of portion 26 of stepped annularly shaped outward appearance disposed on a lower side of the fifth sleeve portion 24; and a sixth sleeve portion 27 of stepped cylindrically shaped outward appearance disposed on a lower side of the coupler case of portion 26. The coupler case of portion 26 couples the fifth sleeve portion 24 and the sixth sleeve portion 27. Note that part of the sixth sleeve portion 27 (a portion thereof disposed within the intermediate lens barrel 22) configures a truncated conical portion 27a which is formed in a downwardly tapering truncated conical shape. Although in the example illustrated, outer diameters of the fifth sleeve portion 24 and the sixth sleeve portion 27 are substantially the same and a minimum outer diameter of the coupler case of portion 26 is shown larger than these outer diameters, outer diameters of the fifth sleeve portion 24, the coupler

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case of portion 26, and the sixth sleeve portion 27 may be configured to be substantially the same.

On an inside of the fourth sleeve portion 23 in the first lens barrel 21, there is housed a light-emitting end LE of the fitted light source LS. Moreover, the fifth sleeve portion 24 in the first lens barrel 21 houses on its inside on the upper side a collector lens (a light-collecting lens) 33. Furthermore, on an inside of a connecting point of the fifth sleeve portion 24 and the coupler case of portion 26, there is housed a fly-eye lens 34.

The sixth sleeve portion 27 in the first lens barrel 21 houses on its inside on the upper side a first relay lens group 35. The first relay lens group 35 has a plurality of lenses that are not joined to each other. For example, in the example illustrated, the first relay lens group 35 comprises: a first lens 35a consisting of a plano-convex lens having its convex surface directed to the lower side in the Z axis direction; and a second lens 35b consisting of a plano-convex lens having its convex surface directed to the upper side in the Z axis direction. These first lens 35a and second lens 35b are disposed so that fellow convex surfaces face each other. The first lens 35a and the second lens 35b may be disposed separated from each other, or may have parts of their convex surfaces in contact with each other.

Note that an inner peripheral surface 27b more to the lower side than a housing place of the first relay lens group 35 of the sixth sleeve portion 27 in the first lens barrel 21 is formed in a tapered shape whose diameter gradually decreases in a downward direction. Moreover, part of an outer peripheral portion of the sixth sleeve portion 27 has formed therein a flange portion 27c of annular shape projecting in an outer peripheral direction.

Further to the lower side than a lower end surface 27d of the flange portion 27c of the sixth sleeve portion 27 in the first lens barrel 21, an outer peripheral portion 27e whose diameter is smaller than a minimum outer diameter of the sixth sleeve portion 27 is passed, and a stepped portion formed still further below is traversed, whereat there is disposed an outer peripheral portion 27f of the truncated conical portion 27a, that is formed in a tapered shape whose diameter gradually decreases in a downward direction.

Note that the fourth sleeve portion 23, fifth sleeve portion 24, coupler case of portion 26, and sixth sleeve portion 27 configuring the first lens barrel 21 are linearly coaxially disposed along the optical axis L1 of the first lens barrel 21. Moreover, a lower end portion of the truncated conical portion 27a of the sixth sleeve portion 27 configures a field stop 36. That is, the field stop 36 in the illumination lens barrel 20 according to the first embodiment is configured by part of the first lens barrel 21 (a lower end portion of the truncated conical portion 27a).

The intermediate lens barrel 22 connects the first lens barrel 21 and the main lens barrel 10 and has an intermediate case of portion 28 of rectangular outward appearance that, in the basic attitude, has formed therein on a side of its end portion separated in the X axis direction from the main lens barrel 10 an inclined wall 28a inclining obliquely downwardly from the upper side toward the main lens barrel 10.

Note that an end portion on a side of the main lens barrel 10 in the X axis direction of the intermediate case of portion 28 is formed as a mating cylindrical portion 28b of cylindrical shape, and is mated with a mating portion 16a which is hole-like, formed in the second casing 16 of the case of main body portion 13. As a result, the intermediate lens barrel 22 is connected to the main lens barrel 10 so as to be capable of revolving around the optical axis L2 orthogonal to the optical axis L1 of the first lens barrel 21, with respect

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to the main lens barrel 10 (refer to FIG. 2). Note that a second relay lens 38 is housed on an inside of the mating cylindrical portion 28b.

Moreover, an upper wall portion 28c of the intermediate case of portion 28 in the basic attitude has formed therein a mating hole 28d with which the sixth sleeve portion 27 of the first lens barrel 21 is mated. That is, the sixth sleeve portion 27 is mated with the intermediate case of portion 28 in such a state that the lower end surface 27d of the flange portion 27c abuts on an outer wall surface of the upper wall portion 28c and the outer peripheral portion 27e catches on an inner peripheral surface on an opening end side of the mating hole 28d.

Note that the sixth sleeve portion 27 is fixed to the intermediate case of portion 28 by the flange portion 27c being fitted to the upper wall portion 28c by fitting bolts 19. As a result, the first lens barrel 21 is fixed after having been positioned with respect to the intermediate lens barrel 22. This configuration results in the first lens barrel 21 being fixed in this way to the intermediate lens barrel 22, so leads to the above-mentioned field stop 36 of the first lens barrel 21 being disposed more inwardly than an outer peripheral surface of the intermediate lens barrel 22 (the outer wall surface of the upper wall portion 28c of the intermediate case of portion 28) is.

Moreover, the intermediate lens barrel 22 has housed therein: a reflecting mirror 37; and a mirror frame member 40 that holds this reflecting mirror 37. Moreover, the intermediate lens barrel 22 is provided with a position adjusting mechanism (not illustrated) for adjusting a position of fitting to the intermediate lens barrel 22 (a position of fitting to the intermediate casing portion 28) of the mirror frame member 40.

FIG. 3 is a schematic side view showing configuration of the mirror frame member 40 of the microscope unit 100. Moreover, FIG. 4 is a schematic perspective view showing configuration of the mirror frame member 40 of the microscope unit 100.

As shown in FIGS. 3 and 4, the mirror frame member 40 is formed by a frame body 49 that, so as to allow the mirror frame member 40 to be housed inside the intermediate case of portion 28, has a cylindrical shape of a kind that one end portion side thereof has been obliquely cut. The frame body 49 of the mirror frame member 40 holds the reflecting mirror 37. This reflecting mirror 37 is formed with an elliptical or polygonal (octagonal, or the like) outer shape, for example, so that its reflecting portion will have an elliptical shape matching a shape of an elliptical opening portion 41 of the frame body 49. Moreover, a portion excluding the above-described reflecting portion, of this reflecting mirror 37 is adhered to the frame body 49 of the mirror frame member 40 via the likes of an unillustrated adhesive, for example.

Moreover, the frame body 49 of the mirror frame member 40 has: a main lens barrel side opening portion 42 formed at an end portion on an opposite side in the X axis direction to the elliptical opening portion 41, of the frame body 49; and a first lens barrel side opening portion 43 formed at a position separated in the Z axis direction from the elliptical opening portion 41, looking from the elliptical opening portion 41. The first lens barrel side opening portion 43 has an opening diameter larger than an outer diameter of the field stop 36 of the truncated conical portion 27a in the sixth sleeve portion 27 of the first lens barrel 21 and larger than an outer diameter of the outer peripheral portion 27f in a certain range on the upper side in the Z axis direction from the field stop 36. As a result, the field stop 36 can be disposed brought as close as is possible to the reflecting mirror 37,

without the frame body 49 of the mirror frame member 40 interfering with the truncated conical portion 27a of the sixth sleeve portion 27.

Note that after the mirror frame member 40 thus configured has been housed inside the intermediate case of portion 28, the mirror frame member 40 has its fitting position adjusted with respect to the intermediate case of portion 28 by the unillustrated position adjusting mechanism. Although illustration of the position adjusting mechanism is omitted, it is configured by a plurality of adjustment screws that have been screwed into a pair of seating countersunk holes drilled in the Y axis direction in the intermediate case of portion 28, for example. After adjustment of the fitting position, the mirror frame member 40 is fixed to the intermediate case of portion 28 by screw fixing, for example.

Note that since the thus formed intermediate lens barrel 22 and the separately assembled first lens barrel 21 can be integrated by a simple structure only of the flange portion 27c being fixed by screw fixing to the intermediate case of portion 28 by the fitting bolt 19 in a state of the truncated conical portion 27a having been inserted in the mating hole 28d, it is possible for the illumination lens barrel 20 to be easily manufactured.

[Optical Systems]

The imaging optical system of the microscope unit 100 according to the present embodiment comprises the objective lens 32 and the imaging lens (tube lens) 17. Moreover, the illumination optical system of the microscope unit 100 according to the present embodiment comprises the collector lens (light-collecting lens) 33, the fly-eye lens 34, the first relay lens group 35, the field stop 36, the reflecting mirror 37, the B/S 18, and the second relay lens 38.

The light-emitting end LE of the light source LS, a position of seating of the objective lens 32 (a pupil of the objective lens 32), and an emitting end surface of the fly-eye lens 34 are in a conjugate relationship. Moreover, these positions are not in a conjugate relationship with the B/S 18 and the reflecting mirror 37. Moreover, the field stop 36, a focal position of the objective lens 32, and a focal position of the imaging lens 17 are in a conjugate relationship.

A light flux (a flux of diverging light rays) from the light-emitting end LE of the light source LS is converted into a collected light flux (a flux of focused light rays) via the collector lens 33, and passes through the fly-eye lens 34 to be divided into the same number as there are a plurality of lenses configuring the fly-eye lens 34. As a result, the emitting end surface of the fly-eye lens 34 has formed therein the same number of light source images as there are a plurality of lenses configuring the fly-eye lens 34 (secondary light sources), and light fluxes (fluxes of diverging light rays) emitted for each of the plurality of lenses configuring the fly-eye lens 34 pass through the first relay lens group 35 to be relayed collected overlapping each other on the field stop 36. Hence, the emitting end surface of the fly-eye lens 34 has formed therein a pseudo surface light source (secondary light source) having substantially uniform light distribution characteristics with little light distribution variation, even when a variety of light sources LS having different light distribution characteristics have been employed.

A light flux (a flux of diverging light rays) that has passed through the field stop 36 along the optical axis L1 and had its range of light stopped down, is reflected so as to become a light flux (a flux of diverging light rays) along the optical axis L2 by the reflecting mirror 37 in the intermediate lens barrel 22, and, having passed through the second relay lens 38 to undergo light collection, is relayed to be irradiated onto

the B/S 18. Note that the B/S 18 guides at least a part of the light flux (the flux of diverging light rays) that has passed through the second relay lens 38 to be incident on the B/S 18 reflecting it to a side of an unillustrated aperture stop and the objective lens 32, and allows to be transmitted through the B/S 18 to a side of the imaging lens 17 at least a part of a light flux from an imaging surface IS that has passed through the objective lens 32, and so on, to be incident on the B/S 18.

[Rotation of Illumination Lens Barrel 20 and Positioning of Reflecting Mirror 37]

As mentioned above, in the microscope unit 100 according to the present embodiment, the illumination lens barrel 20 is configured able to revolve with respect to the main lens barrel 10. Now, in the case of a position with respect to the B/S 18 of the reflecting mirror 37 not having been suitably adjusted, there is a risk that when the illumination lens barrel 20 is rotated with respect to the main lens barrel 10, a center position of the field stop 36 will end up deviating from an optical axis of the illumination optical system. Accordingly, the microscope unit 100 according to the present embodiment is configured to allow the position with respect to the B/S 18 of the reflecting mirror 37 to be adjusted. This point will be described below.

FIGS. 5a and 5b are diagrams for explaining a relationship between attitude of the illumination lens barrel 20 with respect to the main lens barrel 10 of the microscope unit 100 and illumination range. FIGS. 6a, 6b and 6c are diagrams for explaining illumination range in the case of fitting position adjustment having been completed and the case of fitting position adjustment not yet having been performed. Note that in FIGS. 5a, 5b, 6a, 6b and 6c, the outward appearance of the microscope unit 100 differs from that shown in FIGS. 1 and 2. Moreover, description is made here assuming fitting position and fitting angle of the B/S 18 of the main lens barrel 10 to have already been adjusted.

In the microscope unit 100 in the basic attitude with the optical axis ML of the main lens barrel 10 and the optical axis L1 of the first lens barrel 21 of the illumination lens barrel 20 being in the Z axis direction as shown in FIG. 5a, a center of an imaging range 51 in the imaging surface IS and center of an illumination range 52 of light from the light source LS that has passed through the illumination lens barrel 20 are fixed at a regular position centered on the optical axis ML of the main lens barrel 10, that is, at such a position that the center of the imaging range 51 and center of the illumination range 52 will coincide with the optical axis ML, as shown in FIG. 5b.

In the microscope unit 100 according to the present embodiment, it is possible for the illumination lens barrel 20 to be rotated by 90° around the optical axis L2 with respect to the main lens barrel 10, for example, from this basic attitude, as shown in FIG. 6a. As a result, the illumination range 52 rotates by 90° with an image of the imaging range 51 left unchanged.

Now, if, for example, fitting position of the mirror frame member 40 has been adjusted by the above-mentioned position adjusting mechanism, it is possible, as shown in FIG. 6b, for a state where the center of the imaging range 51 and center of the illumination range 52 coincide with the optical axis ML to be maintained, regardless of there having been a change in attitude indicated by the arrow in the drawing.

On the other hand, in the case of fitting position of the mirror frame member 40 not having been adjusted, the center of the illumination range 52 will end up being in a state of having greatly deviated from the center of the

imaging range **51**, as shown in FIG. **6c**, hence illumination distribution by the light source **LS** in a visual field will end up getting ununiform.

In order to avoid such a situation, in the microscope unit **100** according to the present embodiment, a position of fitting to the intermediate lens barrel **22** of the mirror frame member **40** is adjusted via the position adjusting mechanism. Adjustment of fitting position of the mirror frame member **40** is performed in such a manner that, for example, subsequent to the illumination lens barrel **20** having been assembled, above-mentioned adjustment screws are operated, whereby a disposing position in the X axis direction with respect to the intermediate case of portion **28** of the mirror frame member **40** and a disposing position in a rotation direction centered on the X axis direction are adjusted.

Adjustment is performed so that an amount of eccentricity of the optical axis **L2** accompanying a change in attitude of the first lens barrel **21** at a time when the optical axis **L1** of light from the light source **LS** is converted into light of the optical axis **L2** by the reflecting mirror **37**, will fall within a certain range. This makes it possible for attitude with respect to the main lens barrel **10** of the illumination lens barrel **20** to be freely changed, without the illumination range **52** due to the illumination optical system deviating. [Downsizing of Microscope Unit]

Up to now, there has been known a microscope unit comprising: a main lens barrel of an imaging optical system, the main lens barrel being configured capable of being fitted with an imaging sensor and an objective lens; and an illumination lens barrel of an illumination optical system, the illumination lens barrel being connected to the main lens barrel and configured capable of being fitted with a light source. Such a microscope unit is configured so that, for example, a light flux (a flux of diverging light rays) that has been emitted from the light source is collected by a collector lens, passes through a relay lens system and field stop to be reflected by a mirror, and passes through a relay lens system, aperture stop, and condenser lens to illuminate an observation surface.

Such a microscope unit is sometimes capable of being fitted with a variety of light sources. However, it has sometimes been the case that, depending on a kind of light source fitted to the microscope unit, illumination at the observation surface ends up getting ununiform. Accordingly, in the microscope unit **100** according to the present embodiment, the illumination optical system has the fly-eye lens **34** disposed therein, whereby uniformity of illumination at the observation surface is achieved.

However, in the case of such a configuration being adopted, the number of optical components configuring the illumination optical system increases, so that sometimes, total length ends up increasing, and it ends up getting more difficult for downsizing to be achieved.

Accordingly, in the illumination lens barrel **20** of the microscope unit **100** according to the first embodiment, downsizing of the microscope unit is achieved by a method of the kind described below.

[Position Adjustment of Field Stop **36**]

In the microscope unit **100** according to the first embodiment, the field stop **36** is disposed more inwardly than the outer peripheral surface of the intermediate lens barrel **22** is. Hence, it is possible for space inside the first lens barrel **21** to be utilized more effectively compared to in such a case as when, for example, the field stop **36** is disposed more outwardly than the outer peripheral surface of the intermediate lens barrel **22** (for example, more to the upper side than

the lower end surface **27d** of the flange portion **27c** of the sixth sleeve portion **27** in FIG. **1**) is. This makes it possible for total length of the illumination lens barrel **20** to be configured short, whereby downsizing of the illumination lens barrel overall is achieved, and, consequently, downsizing of the microscope unit **100** is achieved.

In the case of downsizing of the microscope unit **100** being achieved by such a configuration, the field stop **36** is desirably disposed brought as close as is possible to a vicinity of the reflecting mirror **37**. For example, as shown in FIG. **1**, the field stop **36** is desirably disposed more inwardly than an inner peripheral surface **28e** of the intermediate lens barrel **22**. However, if the field stop **36** ends up interfering with the light flux of the reflecting mirror **37**, then there ends up occurring deterioration in illumination performance (deterioration in uniformity of brightness or illumination) of the illumination optical system. Hence, as shown in FIG. **1**, for example, the field stop **36** is desirably sufficiently separated from the reflecting mirror **37** to prevent it from interfering with the light flux of the reflecting mirror **37**. Moreover, as mentioned above, the field stop **36** and the focal position of the objective lens **32** are configured to be in a conjugate relationship. Now, if the reflecting mirror **37** is not positioned outside a range of focal depth of the field stop **36**, then there is a risk of dirt or scratches, and so on, on a surface of the reflecting mirror **37** ending up being projected onto the imaging surface **IS**. Hence, the reflecting mirror **37** is desirably provided outside the range of focal depth of the field stop **36**.

Note that the sixth sleeve portion **27** of the illumination lens barrel **20** according to the present embodiment has the truncated conical portion **27a** formed in its lower end portion in FIG. **1**, and a lower end portion of this truncated conical portion **27a** configures the field stop **36**. Such a field stop **36** can be produced by a single chuck by a lathe device, or the like, for example, at a time of the sixth sleeve portion **27** being formed, so is capable of being manufactured with high accuracy while manufacturing man-hours are suppressed. Moreover, it is capable of being realized without components for a separate stop, or the like, being used.

Moreover, as mentioned above, in the microscope unit **100** according to the present embodiment, the illumination lens barrel **20** is capable of being rotated with respect to the main lens barrel **10**. Now, depending on configuration of the field stop, there is a risk that when the illumination lens barrel **20** is rotated with respect to the main lens barrel **10**, a center position of the field stop will end up deviating from the optical axis of the illumination optical system. Now, the field stop **36** according to the present embodiment is configured from part of the sixth sleeve portion **27** housing the first relay lens group **35**. Hence, it is possible for a positional relationship of the center position of the field stop and the optical axis of the illumination optical system to be suitably maintained, even when the illumination lens barrel **20** has been rotated with respect to the main lens barrel **10**.

[Design of First Relay Lens Group **35**]

The first relay lens group **35** according to the present embodiment comprises: the first lens **35a** consisting of a plano-convex lens having its convex surface directed to the lower side in FIG. **1**; and the second lens **35b** consisting of a plano-convex lens having its convex surface directed to the upper side in FIG. **1**. Such a configuration makes it possible for a focal distance from an emitting end of light of the first relay lens group **35** to a position of the field stop **36** to be configured shorter, and for even more downsizing of the microscope unit to thereby be achieved, compared to when, for example, a single relay lens has been disposed. More-

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over, due to there being adopted a plano-convex lens rather than an aspherical lens, it is possible for a reduction in manufacturing costs to be realized. Note that from a viewpoint of downsizing of the illumination lens barrel overall, it is suitable for distance in an optical path from the first relay lens group 35 to the second relay lens 38 to be set so as to be no more than five times an outer diameter of the second relay lens 38, for example. Doing so makes it possible for total length of the illumination lens barrel to be designed shorter.

Second Embodiment

FIGS. 7a and 7b are schematic explanatory diagrams showing configuration of a microscope unit 100 according to a second embodiment. FIG. 7a shows overall configuration of the microscope unit 100. In FIG. 7a, part of the microscope unit 100 is shown as a cross-sectional view, and part of the microscope unit 100 is shown as a transparency view. FIG. 7b shows in enlarged form configuration of part of the microscope unit 100. Note that in the description hereafter, configuring elements the same as or corresponding to those of the first embodiment will be assigned with the same symbols as in the first embodiment, and duplicated descriptions thereof will be omitted.

As shown in FIG. 7a, the microscope unit 100 according to the second embodiment differs from the microscope unit 100 according to the first embodiment in that structure of a field stop 36A in the illumination lens barrel 20 differs from structure of the field stop 36 formed in the truncated conical portion 27a of the sixth sleeve portion 27 of the first lens barrel 21.

That is, in the microscope unit 100 according to the second embodiment, for example, as shown in FIG. 7b, a mirror frame member 40A is housed inside the intermediate lens barrel 22. Moreover, the field stop 36A according to the second embodiment is disposed in the mirror frame member 40A. That is, the field stop 36A is provided as a separate component rather than being formed as part of the sixth sleeve portion 27 in the first lens barrel 21 of the illumination lens barrel 20 according to the first embodiment.

This field stop 36A, which consists of an annular round-shaped flat plate member, is adhered via the likes of an adhesive to an upper end surface of a flange-like fitting portion 43a that is formed on a side of a lower opening end of the first lens barrel side opening portion 43 formed in the frame body 49 of the mirror frame member 40A, in such a manner that the fitting portion 43a projects toward a center of the first lens barrel side opening portion 43. As a result, the field stop 36A is disposed more inwardly than an outer peripheral surface 49a of the frame body 49 of the mirror frame member 40A, similarly to the field stop 36 in the first embodiment.

Note that in the second embodiment, a sixth sleeve portion 27A of the first lens barrel 21 does not have formed therein the truncated conical portion 27a that had been present more to the lower side than the outer peripheral portion 27e had been, hence it is possible for structure of the sixth sleeve portion 27A to be more simply configured. In the microscope unit 100 according to the second embodiment too, downsizing of the illumination lens barrel 20 is realizable, similarly to in the first embodiment.

[Outward Appearance of Microscope Units 100 According to First Embodiment and Second Embodiment]

FIGS. 8a, 8b, 9a, 9b, 9c, 10a, 10b, and 10c are explanatory diagrams showing outward appearance of the above-mentioned microscope units 100. FIG. 8a is a perspective

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view from above of the microscope unit 100. FIG. 8b is a perspective view from below of the microscope unit 100. FIG. 9a is a front view of the microscope unit 100. FIG. 9b is a back view (a rear view) of the microscope unit 100. FIG. 9c is a plan view (a top view) of the microscope unit 100. FIG. 10a is a bottom surface view (a bottom view) of the microscope unit 100. FIG. 10b is a right-side view of the microscope unit 100. FIG. 10c is a left-side view of the microscope unit 100. Note that in FIGS. 8a to 10c, main symbols assigned to each of the previously described configuring elements of the main lens barrel 10 and illumination lens barrel 20 are not shown, and duplicated descriptions thereof will be omitted.

The microscope unit 100 is to be fitted to a microscope to provide an imaging optical system and illumination optical system for observing or imaging an object of an observation surface. As mentioned above, in the microscope unit 100, the illumination lens barrel 20 configuring the illumination optical system of the microscope is configured smaller-sized than the main lens barrel 10 configuring the imaging optical system of the microscope, and, moreover, is integrally connected to the main lens barrel 10 in a manner enabling it to revolve with respect to the main lens barrel 10 such that its attitude freely changes from the basic attitude. The main lens barrel 10 is capable of being fitted with an imaging sensor and objective lens of the microscope. The illumination lens barrel 20, which is capable of being fitted with a variety of light sources, is connected to the main lens barrel 10.

As shown in FIGS. 8a and 8b, the cylindrical first sleeve portion of the main lens barrel 10 has: a device-connecting flange portion 101 of annular shape provided on an upper side of the first sleeve portion; and a first fitting flange portion 102 of rectangular shape provided on a lower side of the first sleeve portion. The cylindrical second sleeve portion of the main lens barrel 10 has: a second fitting flange portion 103 of rectangular shape provided on an upper side of the second sleeve portion; and a third fitting flange portion 104 of rectangular shape provided on a lower side of the second sleeve portion.

The first sleeve portion and second sleeve portion are connected by the first fitting flange portion 102 and second fitting flange portion 103 having their positions aligned, after which, as shown in FIGS. 8a and 9c, places on inner sides of their four corners are each fixed by a bolt 91. The first casing and second casing of the case of main body portion of the main lens barrel 10 are formed in shapes of boxes with chamfered corners, of rectangular shape, having substantially the same outer shapes, as shown in FIGS. 9a, 9b, 10b, and 10c.

The second sleeve portion is connected to the case of main body portion by the third fitting flange portion 104 having its position aligned with an upper portion of the first casing of the case of main body portion, after which, as shown in FIGS. 8a, 9a, 9b, 10b, and 10c, places on inner sides of its four corners are each fixed by a bolt 92.

The cylindrical third sleeve portion of the main lens barrel 10 has a fourth fitting flange portion 105 of annular shape provided on an upper side of the third sleeve portion. The third sleeve portion is connected to the case of main body portion by having its position aligned with a lower portion of the second casing of the case of main body portion, after which, as shown in FIGS. 8b, 9a, 9b, and 10a to 10c, a certain four places thereof are each fixed by a bolt 93. Note that the first casing and second casing of the case of main body portion are integrally configured by being fixed by a

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certain number of bolts **94** from a lower side of the second casing, as shown in FIG. **10a**.

As shown in FIGS. **8a** and **8b**, the first lens barrel of the illumination lens barrel **20** is formed in a cylindrical shape, and the intermediate lens barrel of the illumination lens barrel **20** is formed in a rectangular tubular shape having an inclined portion. As shown in FIGS. **8a**, **8b**, **9a** to **9c**, and **10b**, an outer diameter of the fourth sleeve portion of the first lens barrel is formed smaller than an outer diameter of the fifth sleeve portion of the first lens barrel, and the outer diameter of the fifth sleeve portion is formed smaller than an outer diameter of the coupler case of portion **26** of the first lens barrel. Moreover, an outer diameter of the sixth sleeve portion of the first lens barrel is formed with substantially the same diameter as the outer diameter of the fifth sleeve portion.

As mentioned above, the first lens barrel is fixed by the fitting bolts **19** to the intermediate lens barrel, while, as shown in FIGS. **8a**, **8b**, **9a**, **9b**, **10a**, and **10b**, the intermediate lens barrel is connected to the main lens barrel by a pair of fifth fitting flange portions **106** provided on a second casing side of the case of main body portion of the main lens barrel being fixed by a certain number of bolts **95**. Moreover, the inclined portion of the intermediate lens barrel is formed in a freely attachable/detachable manner in a rounded-cornered rectangular shape, and is fixed to the intermediate lens barrel by a pair of diagonally-aligned bolts **96**, as shown in FIGS. **8b**, **10a**, and **10b**.

OTHER EMBODIMENTS

Each portion of the microscope unit **100** in each of the above-mentioned embodiments is capable of having its outer shape, dimensions, and so on, appropriately changed. For example, it is also possible for the case of main body portion **13** of the main lens barrel **10** to be formed in a cylindrical shape, and for an entirety from the first sleeve portion **11** to the third sleeve portion **14** to be formed in a cylindrical shape. Moreover, it is also possible for the first sleeve portion **11**, second sleeve portion **12**, and third sleeve portion **14** to be formed in a rectangular shape, and for the entirety from the first sleeve portion **11** to the third sleeve portion **14** to be formed in a rectangular (rectangular tubular) shape. Furthermore, it is also possible for outer diameters of the first through third sleeve portions **11**, **12**, **14** to be made different from what was mentioned above. The above changes may be applied to the illumination lens barrel **20** too. Moreover, portions of the microscope unit **100** exemplified as configurations consisting of a plurality of members may be configured from a single member, or portions of the microscope unit **100** exemplified as configurations consisting of a single member may be configured from a plurality of members.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms: furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

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What is claimed is:

1. A microscope unit comprising:

a main lens barrel of an imaging optical system, the main lens barrel extending along a first optical axis and being configured to be capable of being fitted with an imaging sensor and an objective lens;

an intermediate lens barrel extending along a second optical axis orthogonal to the first optical axis, and including an end in a direction in which the second optical axis extends that is connected to an outer peripheral surface of the main lens barrel; and

a first lens barrel extending along a third optical axis orthogonal to the second optical axis, including an end in a direction in which the third optical axis extends that is connected to an outer peripheral surface of the intermediate lens barrel, and that is configured to be capable of being fitted with a light source, wherein

the end of the first lens barrel in the direction in which the third optical axis extends comprises a truncated conical portion formed in a truncated conical shape tapering toward the intermediate lens barrel,

the truncated conical portion is inserted more inwardly than an outer peripheral surface of the intermediate lens barrel is, and configures a field stop of light irradiated from the light source.

2. The microscope unit according to claim 1, wherein the intermediate lens barrel is inserted more inwardly than an inner peripheral surface of the intermediate lens barrel is.

3. The microscope unit according to claim 2, wherein the intermediate lens barrel and the first lens barrel are configured to be capable of revolving around the second optical axis with respect to the main lens barrel.

4. The microscope unit according to claim 3, having: a reflecting mirror housed on an inside of the intermediate lens barrel and reflecting light from the first lens barrel toward the main lens barrel;

a mirror frame member holding the reflecting mirror; and a position adjusting mechanism for adjusting a position of fitting to the intermediate lens barrel of the mirror frame member,

wherein the intermediate lens barrel is inserted more inwardly than an outer peripheral surface of the mirror frame member is.

5. The microscope unit according to claim 4, further comprising an illumination optical system, the illumination optical system comprising:

a collector lens that collects light that has been irradiated from the light source;

a fly-eye lens allowing to be transmitted therethrough light from the collector lens;

a first relay lens that relays light from the fly-eye lens; the field stop that stops down a range of light from the first relay lens;

the reflecting mirror that reflects light from the first relay lens toward the main lens barrel;

a beam splitter provided on an optical axis of the main lens barrel, the beam splitter guiding at least a part of light incident thereon to the objective lens and allowing to be transmitted therethrough to a side of the imaging sensor at least a part of light incident thereon from the objective lens; and

a second relay lens that relays to the beam splitter light that has been reflected by the reflecting mirror, the first lens barrel houses the collector lens, the fly-eye lens, and the first relay lens,

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the intermediate lens barrel houses the second relay lens,
and
the main lens barrel houses the beam splitter.

6. The microscope unit according to claim 2, having:
a reflecting mirror housed on an inside of the intermediate
lens barrel and reflecting light from the first lens barrel
toward the main lens barrel;
a mirror frame member holding the reflecting mirror; and
a position adjusting mechanism for adjusting a position of
fitting to the intermediate lens barrel of the mirror
frame member,
wherein the intermediate lens barrel is inserted more
inwardly than an outer peripheral surface of the mirror
frame member is.

7. The microscope unit according to claim 6, further
comprising an illumination optical system, the illumination
optical system comprising:
a collector lens that collects light that has been irradiated
from the light source;
a fly-eye lens allowing to be transmitted therethrough
light from the collector lens;
a first relay lens that relays light from the fly-eye lens;
the field stop that stops down a range of light from the first
relay lens;
the reflecting mirror that reflects light from the first relay
lens toward the main lens barrel;
a beam splitter provided on an optical axis of the main
lens barrel, the beam splitter guiding at least a part of
light incident thereon to the objective lens and allowing
to be transmitted therethrough to a side of the imaging
sensor at least a part of light incident thereon from the
objective lens; and
a second relay lens that relays light to the beam splitter
that has been reflected by the reflecting mirror,
the first lens barrel houses the collector lens, the fly-eye
lens, and the first relay lens,
the intermediate lens barrel houses the second relay lens,
and
the main lens barrel houses the beam splitter.

8. The microscope unit according to claim 1, wherein
the intermediate lens barrel and the first lens barrel are
configured to be capable of revolving around the sec-
ond optical axis with respect to the main lens barrel.

9. The microscope unit according to claim 8, having:
a reflecting mirror housed on an inside of the intermediate
lens barrel and reflecting light from the first lens barrel
toward the main lens barrel;
a mirror frame member holding the reflecting mirror; and
a position adjusting mechanism for adjusting a position of
fitting to the intermediate lens barrel of the mirror
frame member,
wherein the intermediate lens barrel is inserted more
inwardly than an outer peripheral surface of the mirror
frame member is.

10. The microscope unit according to claim 9, further
comprising an illumination optical system, the illumination
optical system comprising:
a collector lens that collects light that has been irradiated
from the light source;

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a fly-eye lens allowing to be transmitted therethrough
light from the collector lens;
a first relay lens that relays light from the fly-eye lens;
the field stop that stops down a range of light from the first
relay lens;
the reflecting mirror that reflects light from the first relay
lens toward the main lens barrel;
a beam splitter provided on an optical axis of the main
lens barrel, the beam splitter guiding at least a part of
light incident thereon to the objective lens and allowing
to be transmitted therethrough to a side of the imaging
sensor at least a part of light incident thereon from the
objective lens; and
a second relay lens that relays light to the beam splitter
that has been reflected by the reflecting mirror,
the first lens barrel houses the collector lens, the fly-eye
lens, and the first relay lens,
the intermediate lens barrel houses the second relay lens,
and
the main lens barrel houses the beam splitter.

11. The microscope unit according to claim 1, having:
a reflecting mirror housed on an inside of the intermediate
lens barrel and reflecting light from the first lens barrel
toward the main lens barrel;
a mirror frame member holding the reflecting mirror; and
a position adjusting mechanism for adjusting a position of
fitting to the intermediate lens barrel of the mirror
frame member,
wherein the intermediate lens barrel is inserted more
inwardly than an outer peripheral surface of the mirror
frame member is.

12. The microscope unit according to claim 11, further
comprising an illumination optical system, the illumination
optical system comprising:
a collector lens that collects light that has been irradiated
from the light source;
a fly-eye lens allowing to be transmitted therethrough
light from the collector lens;
a first relay lens that relays light from the fly-eye lens;
the field stop that stops down a range of light from the first
relay lens;
the reflecting mirror that reflects light from the first relay
lens toward the main lens barrel;
a beam splitter provided on an optical axis of the main
lens barrel, the beam splitter guiding at least a part of
light incident thereon to the objective lens and allowing
to be transmitted therethrough to a side of the imaging
sensor at least a part of light incident thereon from the
objective lens; and
a second relay lens that relays light to the beam splitter
that has been reflected by the reflecting mirror,
the first lens barrel houses the collector lens, the fly-eye
lens, and the first relay lens,
the intermediate lens barrel houses the second relay lens,
and
the main lens barrel houses the beam splitter.

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